

CNN vs SincNet for Speech Emotion Recognition and Speaker Identification on RAVDESS

Azra Acil

April 30, 2026

1 Introduction

This project investigates two deep learning approaches for Speech Emotion Recognition (SER) and Speaker Identification (SID). These tasks are suitable for combined modelling because they share audio characteristics such as pitch and energy. The main idea is that speaker identity acts as a supporting feature for SER. All models were evaluated on the RAVDESS dataset.

Two main architectures were implemented from scratch using PyTorch. The first uses log-mel spectrograms with a 2D CNN for emotion recognition and a 1D CNN-BiLSTM for speaker identification. The second is based on the SincNet architecture (Ravanelli and Bengio, 2018), which works directly on raw audio waveforms using parametrised sinc functions as bandpass filters.

A multi-task model was also built with a shared encoder. Joint modelling had opposite effects: it slightly hurt the CNN’s emotion accuracy but substantially improved SincNet’s, suggesting speaker identity helps emotion recognition more when working with raw audio.

2 Dataset and Preprocessing

All experiments use the RAVDESS dataset (Livingstone and Russo, 2018), which contains 1440 audio recordings from 24 actors (12 male, 12 female) expressing eight emotions. To avoid acoustic confusion between similar classes, four emotions (calm, fearful, disgusted, surprised) were excluded, leaving four (neutral, happy, sad, angry) and 672 samples. The neutral class has only one intensity level, giving 96 samples against 192 for each other emotion; inverse-frequency class weights handle this imbalance during emotion training.

Emotion recognition uses a speaker-independent split (actors 3, 4 for validation; 23, 24 for test) so the model is evaluated on entirely unseen voices. Speaker identification needs all 24 speakers in training, so a random per-speaker split (70/15/15) is used for closed-set identification. The combined multi-task models use the speaker-independent split so emotion results stay comparable; the speaker head only sees 20 of 24 speakers and acts as an auxiliary signal.

The two architectures need different inputs. CNN models use audio resampled to 16 kHz, padded or trimmed to 3 seconds, and converted into 128-band log-mel spectrograms. SincNet uses raw 16 kHz waveforms; following the original paper, training uses random 200 ms crops (3200 samples), and evaluation averages predictions over non-overlapping chunks of the full audio.

3 Mel-Spectrogram Pipeline

The first approach converts audio into log-mel spectrograms, producing (1, 128, 94) tensors that are treated as single-channel images. Figure 1 shows that happy and angry have visibly higher energy across frequency bands than neutral and sad, which explains why these pairs are difficult to distinguish. Both models in this section were implemented from scratch in PyTorch.

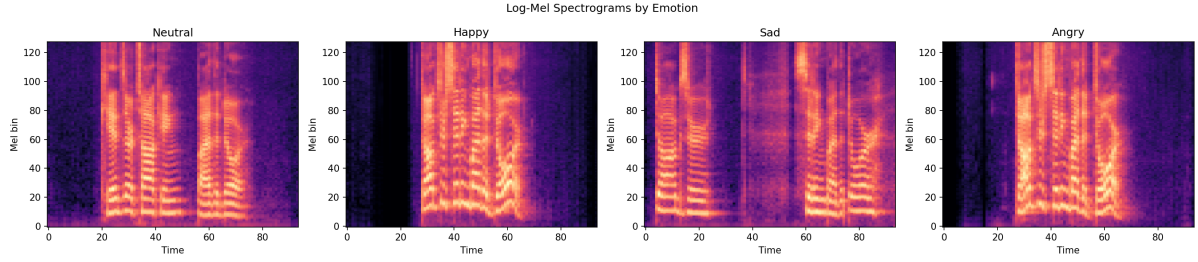


Figure 1: Example log-mel spectrograms for the four emotion classes. Happy and angry display similar high-energy patterns across frequency bands, while neutral and sad show lower overall energy.

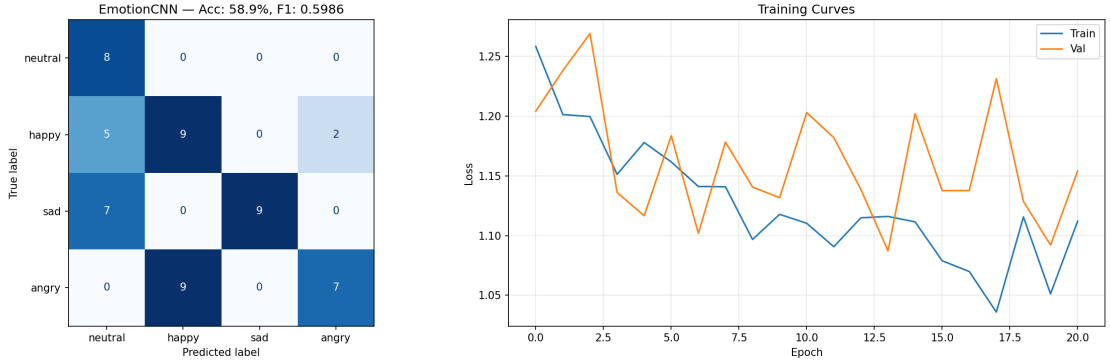


Figure 2: EmotionCNN results. Left: confusion matrix on the held-out test speakers. Right: training and validation loss curves; validation loss remains unstable across epochs, reflecting the difficulty of speaker-independent emotion recognition.

3.1 Emotion Recognition: EmotionCNN

The emotion model is a 2D CNN with five convolutional blocks with filters of (32, 64, 128, 256, 256). Each block uses Conv2D, batch normalisation, ReLU, and Dropout2d, with max pooling while the fifth block omits pooling because the feature maps are already small. The output passes through global average pooling and fully connected layers (256→128→4).

The first version with standard cross-entropy reached only 55.4% accuracy and failed on neutral speech. Adding inverse-frequency class weights fixed the imbalance. Most augmentations (SpecAugment, white noise, pitch shifting) had a negative impact on performance because the dataset is too small, while mixup consistently helped. Hyperparameters were selected through a grid search over 96 configurations, with different dropout, learning rate, augmentation type, scheduler, and mixup. The best validation result (67.9%) used dropout=0.1, lr=0.001, cosine annealing, and mixup. The final model was trained with Adam (lr=0.001), cosine annealing over 30 epochs, and early stopping with patience 7.

On the test set, the final model achieved **58.9% accuracy and 0.599 weighted F1** (Figure 2). The gap between validation (67.9%) and test (58.9%) reflects the challenge of generalising to unseen speakers. The confusion matrix shows that neutral is the best class (8/8 correct), while happy is most confused (5/16 misclassified as neutral), and angry is also problematic (9/16 predicted as happy). This pattern fits with the spectrograms in Figure 1, where happy and angry share similar high-energy characteristics.

3.2 Speaker Identification: SpeakerNet

Speaker identification requires capturing how someone sounds over time rather than a single spectral snapshot, a recurrent architecture was used. Two 1D convolutional blocks (Conv1d, batch normalisation, ReLU, max pooling, dropout) extract local features along the frequency axis. These features pass into a single-layer bidirectional LSTM (hidden size 128) with a soft attention mechanism, which weights the

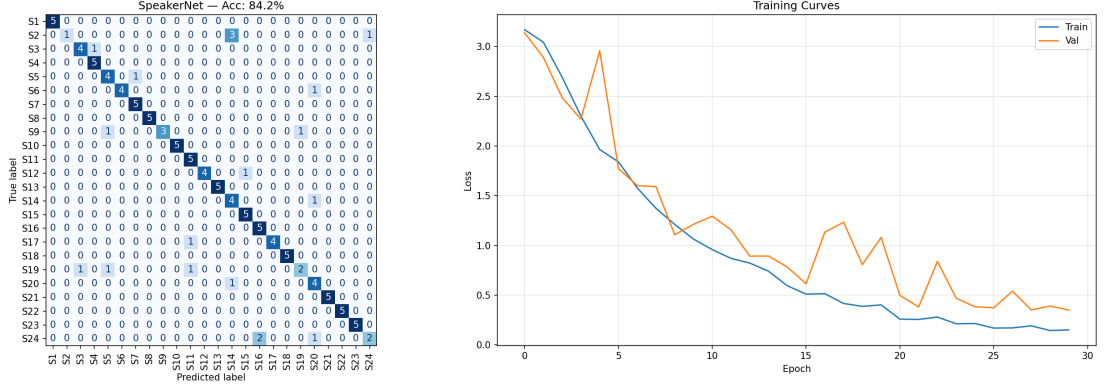


Figure 3: SpeakerNet results. Left: confusion matrix on the 24-speaker closed-set test split. Right: training and validation loss curves.

most speaker-discriminative frames. The attended output goes through fully connected layers to classify 24 speakers.

Hyperparameters were selected through a grid search over 16 configurations, different dropout, learning rate, LSTM hidden size, and scheduler. The best result (81.7%) used dropout=0.1, lr=0.0003, hidden size 128, and ReduceLROnPlateau. Unlike the emotion model, standard cross-entropy loss was used since speakers are equally represented. The final model was trained with Adam (lr=0.0003), ReduceLROnPlateau, and early stopping with patience 7.

The final model achieved **84.2% accuracy** and **0.831 weighted F1** on the closed-set test split (Figure 3). The confusion matrix has a clear diagonal with a few consistent errors: S2 is mostly predicted as S14, and S24 is confused with S16. Almost all errors (17/19) occur between same-gender speakers who share similar pitch ranges and formant structures.

4 Raw-Waveform Pipeline (SincNet)

The second approach works directly on raw 16 kHz audio without mel-spectrograms. SincNet uses parametrised sinc functions as bandpass filters where only two cutoff frequencies are learned per filter. Both models in this section were implemented from scratch in PyTorch, except for the SincConv layer, which was adapted from the official SincNet repository (Ravanelli and Bengio, 2018).

4.1 Emotion Recognition: EmotionSincNet

The encoder uses a SincConv layer with 80 filters of kernel size 251 (initialised on the mel scale, Hamming-windowed), followed by absolute-value activation, max pooling by 3, layer normalisation, and leaky ReLU. Two additional Conv1d blocks (80→60 and 60→60, kernel 5, no padding) follow the same pattern. The output is flattened into a 6420-dimensional vector and passed through LayerNorm → FC(256) → FC(256) → FC(4), giving 1.98M parameters.

This design required iteration. Early versions using global average pooling, instance normalisation, and BiLSTM heads performed poorly (48.2%). Improvement came from following the original paper: valid convolution, flattening instead of pooling, proper layer normalisation, and random 200 ms cropping. The final configuration used dropout=0.2, lr=0.001, no augmentation, and no scheduler. Training used RMSprop following the paper, with inverse-frequency class weights.

The model achieved **50.0% accuracy** and **0.498 weighted F1** after 18 epochs (Figure 4), 8.9 points below EmotionCNN. The confusion matrix shows that angry is the hardest class, with 9/16 misclassified as happy; the same high-energy confusion seen in EmotionCNN, but stronger here. The mel-spectrogram provides a more structured input for emotion recognition on this small dataset.

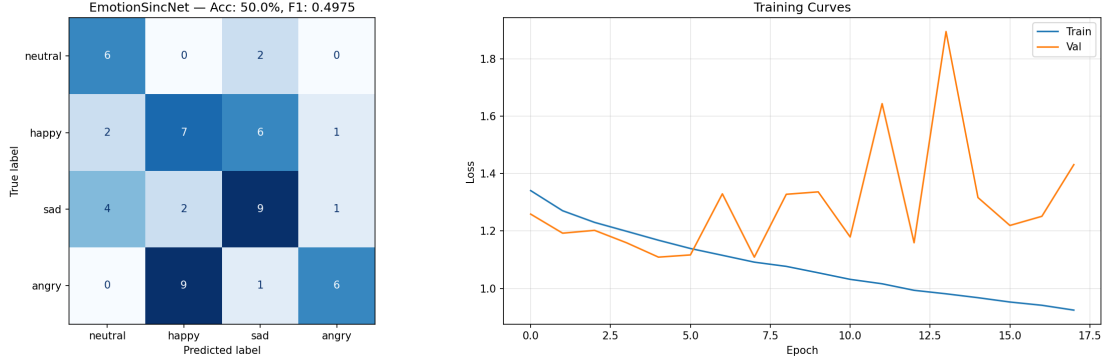


Figure 4: EmotionSincNet results. Left: confusion matrix on the held-out test speakers. Right: training and validation loss curves; validation loss diverges after epoch 5 and spikes above 1.8, indicating overfitting on the limited training set.

4.2 Speaker Identification: SpeakerSincNet

For speaker identification, the same encoder is used as in EmotionSincNet, but with wider FC heads: LayerNorm $\rightarrow$ FC(512) $\rightarrow$ FC(512) $\rightarrow$ FC(24), giving 3.83M parameters. The larger heads provide more capacity for distinguishing 24 speakers. Training used RMSprop with the same settings as EmotionSincNet, dropout=0.2, lr=0.001, and standard cross-entropy loss.

The model was trained for 100 epochs and achieved **93.3% accuracy and 0.930 weighted F1** on the closed-set test split (Figure 5), 9.1 points above SpeakerNet. The confusion matrix is nearly perfectly diagonal, with the few errors concentrating on specific speakers (S20: only 2 of 5 correct) rather than spreading across same-gender pairs as in SpeakerNet.

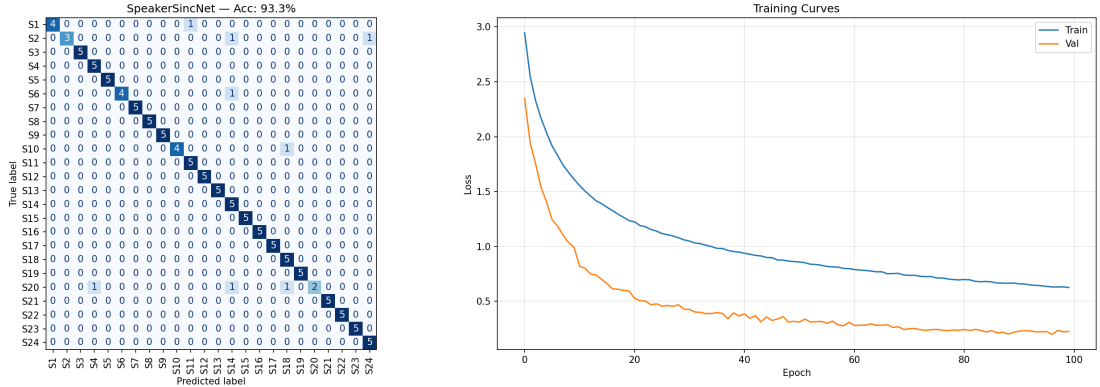


Figure 5: SpeakerSincNet results. Left: confusion matrix on the 24-speaker closed-set test split. Right: training and validation loss curves showing smooth convergence over 100 epochs.

5 Combined System: Multi-Task Learning

Both pipelines were extended into multi-task models by sharing the encoder and attaching two task-specific heads. The total loss is $\mathcal{L} = \mathcal{L}_{\text{emotion}} + \lambda \cdot \mathcal{L}_{\text{speaker}}$, where λ controls the speaker task’s contribution. The idea is that learning to identify the speaker might force the encoder to capture richer features (pitch, formants, speaking style) that also help with emotion recognition. Both architectures were tested with $\lambda \in \{0.0, 0.1, 0.3, 0.5, 0.7, 1.0\}$.

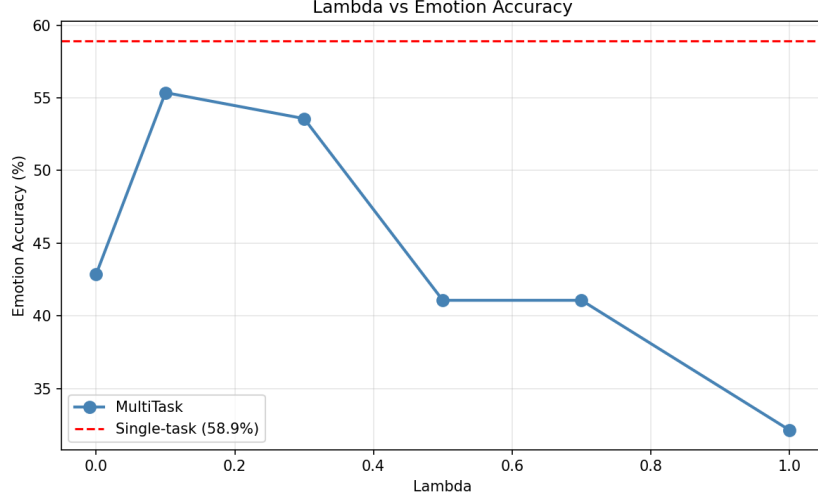


Figure 6: CNN multi-task: effect of speaker loss weight λ on emotion accuracy. The dashed line shows the single-task EmotionCNN baseline (58.9%). All λ values fall below the baseline.

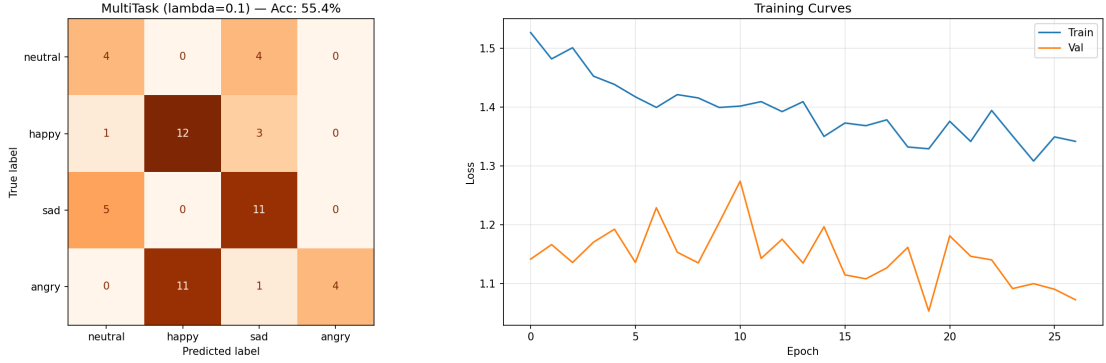


Figure 7: CNN multi-task results at $\lambda = 0.1$. Left: confusion matrix. Right: training and validation loss curves.

5.1 CNN Multi-Task

The MultiTaskNet shares the five-block 2D CNN encoder from EmotionCNN. The emotion head uses global average pooling followed by FC layers (256→128→4), and the speaker head uses a BiLSTM with attention, similar to SpeakerNet.

The lambda sweep results are shown in Figure 6. The best emotion accuracy was 55.4% at $\lambda = 0.1$, still 3.6 points below standalone EmotionCNN (58.9%). At $\lambda = 0.0$ accuracy drops to 42.9% because the shared encoder has less capacity than the dedicated one. Higher λ values reduce performance, reaching 32.1% at $\lambda = 1.0$.

The confusion matrix at $\lambda = 0.1$ (Figure 7) shows a mixed pattern: high-energy emotions improve (happy 12/16 vs 9/16, sad 11/16 vs 9/16), but low-energy classes get worse (neutral 4/8 vs 8/8, angry 4/16 vs 7/16). The speaker signal helps with dynamic emotions but causes confusion in stable ones.

5.2 SincNet Multi-Task

The MultiTaskSincNet shares the SincNet encoder from Section 4.1, with the emotion head (FC 256→256→4) and speaker head (FC 512→512→24) branching from the same 6420-dimensional representation.

The lambda sweep showed a very different pattern, shown in Figure 8. At $\lambda = 0.0$ (emotion only),

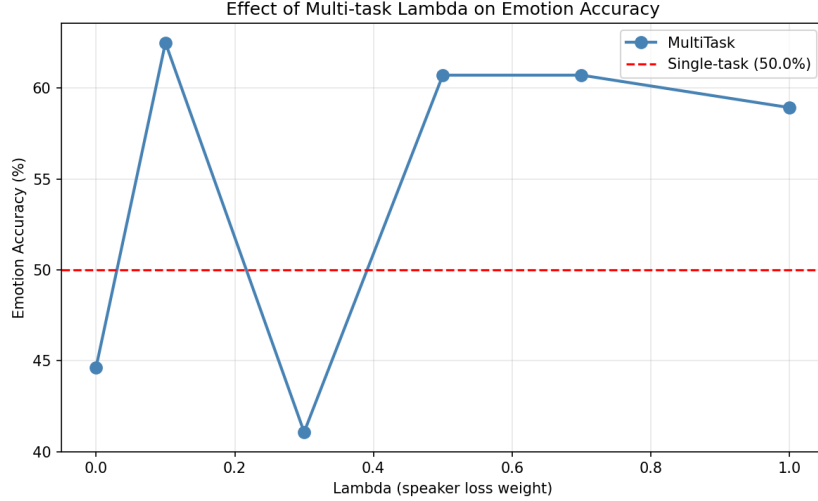


Figure 8: SincNet multi-task: effect of speaker loss weight λ on emotion accuracy. The dashed line shows the single-task EmotionSincNet baseline (50.0%). Almost all λ values land above the baseline, with $\lambda = 0.1$ giving the best result at 62.5%.

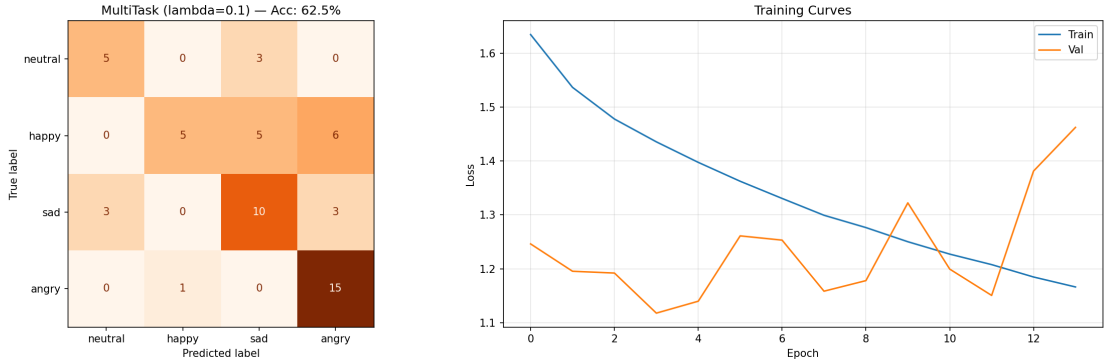


Figure 9: SincNet multi-task results at $\lambda = 0.1$. Left: confusion matrix — angry recognition jumps to 15/16 correct. Right: training and validation loss curves.

accuracy was just 44.6%. Adding the speaker signal consistently improved performance: $\lambda = 0.1$ achieved the best result at 62.5%, and even $\lambda = 1.0$ reached 58.9%, well above the standalone EmotionSincNet (50.0%). The only exception is $\lambda = 0.3$, which dropped to 41.1%, possibly due to a bad gradient interaction at that specific weighting.

The confusion matrix at $\lambda = 0.1$ (Figure 9) shows where the gain comes from: angry recognition jumps from 6/16 (in EmotionSincNet) to **15/16** correct, where a perfect score. Sad also improves slightly (10/16 vs 9/16). Happy gets slightly worse (5/16 vs 7/16) and neutral stays similar (5/8 vs 6/8). The significant improvement in angry which was the worst class in the single-task SincNet, accounts for the most of the +12.5 point improvement.

6 Case Study

For the case study, I picked one held-out test sample to see how all four models behave on the same audio: Actor 24 (female) saying “dogs are sitting by the door“ angrily. Due to the splits, this actor was unseen by the emotion models but seen by the speaker models. Figure 10 shows the waveform and the log-mel spectrogram.

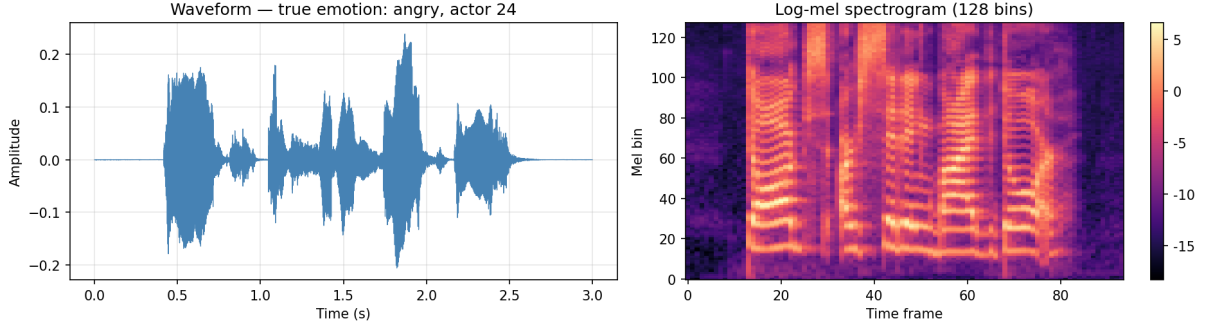


Figure 10: Case study sample. Left: 3s waveform of Actor 24 saying “*dogs are sitting by the door*” with angry intent. Right: corresponding 128-bin log-mel spectrogram.

Table 1: Per-model predictions on the case study sample. True labels: emotion = angry, speaker = Actor 24.

Model	Prediction	Outcome
EmotionCNN (mel)	angry	correct
EmotionSincNet (raw)	happy	wrong
SpeakerNet (mel)	Actor 24	correct
SpeakerSincNet (raw)	Actor 24	correct

Predictions are summarised in Table 1. Both speaker models correctly identify Actor 24, in line with their strong overall accuracy. The emotion models disagree: EmotionCNN classifies the sample as angry (correct), while EmotionSincNet predicts happy (wrong). This is the same high-energy confusion seen in Section 4.1.

7 Discussion and Conclusion

The aim of this work was to see whether multi-task learning could help with speech emotion recognition. As single-task models, the mel-CNN got 58.9% on emotion and 84.2% on speaker, while SincNet got 50.0% on emotion and 93.3% on speaker. Each architecture is clearly better at one task: the mel-spectrogram works well for emotion recognition, while SincNet’s filters are good at picking up speaker features like pitch and formants.

The multi-task experiments gave the most interesting result. Adding a speaker loss made the CNN’s emotion accuracy worse at every λ , but made SincNet’s better at almost every λ , with a +12.5 point gain at $\lambda = 0.1$. The multi-task SincNet (62.5%) even did better than the standalone EmotionCNN (58.9%). The most likely reason is that learning from raw audio is harder, and the speaker task helps push the sinc filters toward useful frequency regions (pitch, formants) that also help with emotion. The mel-CNN doesn’t need this help because the mel-spectrogram already gives the model these features.

Limitations

The dataset is small (672 samples) and the test sets are even smaller (56 emotion, 120 speaker), so the results might change a bit with a different random seed. I also didn’t tune the SincNet emotion head as much as the CNN, so the gap between them might be smaller in practice.

Conclusion

The main thing I found is that multi-task learning works in opposite ways for the two architectures: the same speaker loss makes the CNN worse by 3.6 points but makes SincNet better by 12.5 points. The multi-task SincNet ended up giving the best emotion result of the project (62.5%), which is exactly what I was hoping for when I added the multi-task setup.

References

- [1] Ravanelli, M. and Bengio, Y. (2018). Speaker recognition from raw waveform with SincNet. *IEEE Spoken Language Technology Workshop (SLT)*.
- [2] Livingstone, S. R. and Russo, F. A. (2018). The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS). *PLoS ONE*, 13(5), e0196391.